

Parameter-Efficient Adaptation of SAM 2 for Sentinel-1 SAR Flood Inundation Mapping in the Indo-Gangetic Region

Part 2: Results and Diagnosis

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Headline results

Method	Sen1F11 test	Bolivia	Pak-S1F11	Pak-2022
U-Net baseline	0.601	0.684	0.271	0.672
Hiera-B+ + AdaptFormer	0.634	0.458	0.254	0.090
Hiera-B+ + LoRA	0.599	0.635	0.255	0.140
Hiera-B+ + DoRA	0.602	0.637	0.254	0.132

- **In-distribution:** AdaptFormer wins (0.634), beating U-Net at a fraction of trainable parameters.
- **Bolivia held-out:** DoRA matches U-Net (0.637 vs 0.684).
- **Pakistan-2022:** every SAM-family cell collapses to 0.09–0.14, while U-Net retains 0.672.

The Pakistan-2022 collapse: not a backbone or adapter problem

- Every SAM-family cell collapses, regardless of:
 - Backbone (ViT-Base, ViT-Large, ViT-Huge, Hiera-Base-Plus, Hiera-Large — all collapse)
 - PEFT method (LoRA, DoRA, Conv-LoRA, AdaptFormer — all collapse)
 - Random seed
- U-Net on the same training data does not collapse. So the collapse is *specific to the adapted SAM-family models*, not a property of the dataset.
- **Question:** what is the SAM family seeing that the U-Net is not?

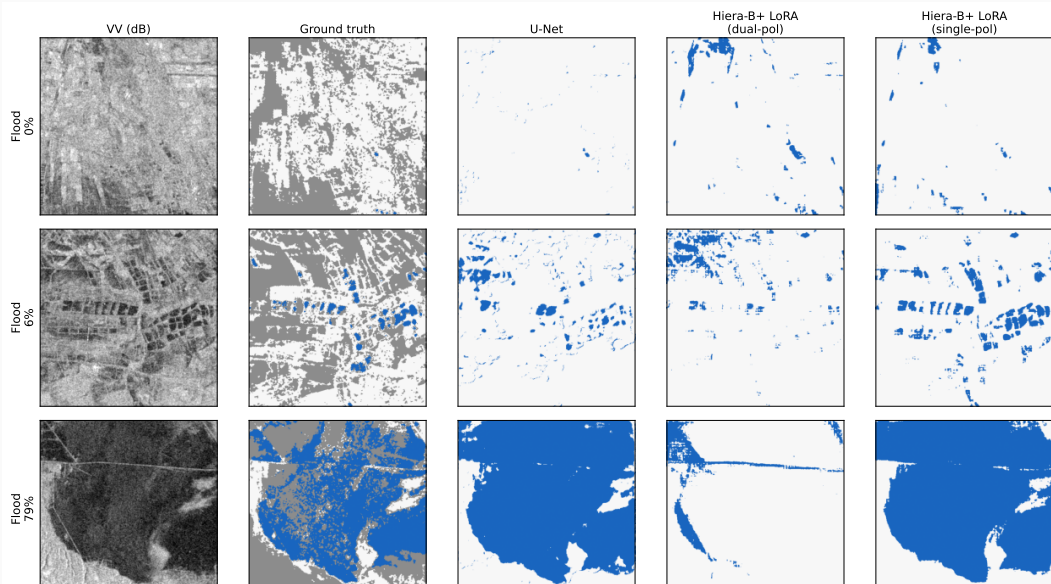
The polarimetric ablation: cross-polarization is the culprit

Pakistan-2022 IoU	Dual-polarization	Single-polarization (VV only)
Hiera-B+ Conv-LoRA	0.113	0.632 ± 0.057
Hiera-B+ LoRA	0.144	0.631 ± 0.050
Hiera-B+ DoRA	0.131	0.658 ± 0.014
Hiera-B+ AdaptFormer	0.092	0.556 ± 0.050
<i>U-Net reference</i>		0.672

Drop the cross-polarization (VH) channel and the collapse vanishes.

- Single-polarization Pakistan-2022 IoU jumps to 0.56–0.66 across three seeds.
- Within 0.01–0.12 of the U-Net baseline.
- **Three-seed evidence; no overlap with the dual-pol band.**

Qualitative comparison: same chip, three predictions



Why? The transformer amplifies cross-polarization distribution shift

- The cross-polarization channel *helps* in-distribution. Its marginal statistics on Pakistan-2022 differ from Sen1Floods11.
- Possible drivers: different scattering geometry, different surface conditions, different acquisition mode parameters.
- The SAM transformer's global attention *amplifies* this distribution shift.
- The U-Net's local receptive fields *tolerate* it.

Operational recommendation: for new regions or events, abstain from the cross-polarization channel. Use VV only.

Negative finding: MC dropout is not per-pixel useful on OOD

Aggregate calibration is fine:

- $ECE = 0.053$ on in-distribution test (at the well-calibrated threshold).
- $ECE = 0.067$ on Pakistan-2022 (essentially unchanged despite the IoU collapse).

But pointwise selective-prediction curves are flat:

- Abstaining on the most uncertain pixels does *not* improve IoU on the remaining ones.
- Aggregate calibration $\nRightarrow$ per-pixel difficulty ranking on OOD events.
- Contradicts the implicit assumption in prior SAR-flood uncertainty work.

Three gaps, three closures

1. **PEFT comparison:** controlled head-to-head; accuracy-versus-generalization trade-off documented. AdaptFormer maximizes in-distribution IoU; LoRA family is more generalization-robust.
2. **Calibration vs operational utility:** substantive negative finding. Aggregate ECE does not imply per-pixel uncertainty utility on OOD.
3. **Public OOD test set:** Pakistan-2022 (39 chips, native 10 m) released with acquisition pipeline.

Bonus diagnostic finding: cross-polarization is the operative OOD failure mode for SAM-family adapters.

Honest limitations:

- Pakistan-2022 evaluation rests on 39 chips and TU Wien labels derived from a Sentinel-1 model.
- Stretch backbones evaluated only under Conv-LoRA.
- Bangladesh-Sylhet construction pipeline is released, chips are not yet built or evaluated.

Future work:

- Empirical evaluation on the Bangladesh-Sylhet test set.
- Characterize the cross-polarization shift directly (incidence angle, acquisition mode).
- Sharper uncertainty estimators: deep ensembles, learned reject heads.